

CSL7870: TASK-11 — GraphSAGE and GAT implementation

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1 Introduction

I implement GraphSAGE and GAT for semi-supervised node classification on the Cora citation network. I keep the depth fixed at two layers for all models. I compare GraphSAGE aggregators (mean, max, LSTM) and vary GAT heads $\{1, 2, 5\}$. I use the Planetoid split and report test accuracy at the epoch that achieves the best validation accuracy. I also include loss and accuracy curves, confusion matrices, and per-class precision, recall, and F1. When the LSTM aggregator is unstable on the 1,433-dimensional input features, I use an autoencoder to produce a 64,128 and 256 -dimensional variant and rerun the experiments.

2 Dataset: Cora

Cora is a citation graph where nodes are papers and edges are citation links. Each node has a sparse bag-of-words feature vector and one of seven topic labels.

2.1 Summary

- **Nodes:** 2,708
- **Features per node:** 1,433
- **Classes:** 7
- **Split (Planetoid):** train 140, validation 500, test 1,000

2.2 Graph statistics

- **Average degree:** 3.90
- **Median degree:** 3
- **Maximum degree:** 168

2.3 Class distribution

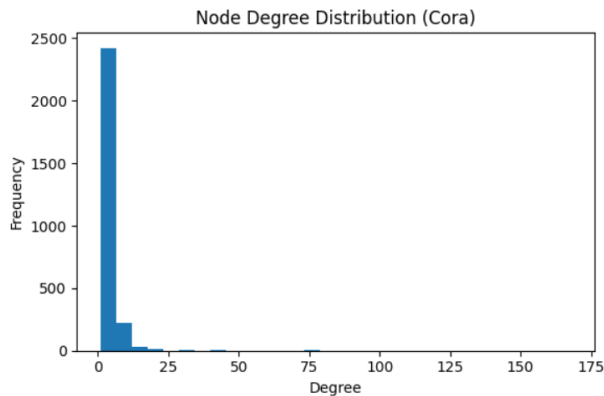
Table 1: Class distribution on the node set.

Class ID	Count	Percent (%)
0	351	12.96
1	217	8.01
2	418	15.44
3	818	30.21
4	426	15.73
5	298	11.00
6	180	6.65

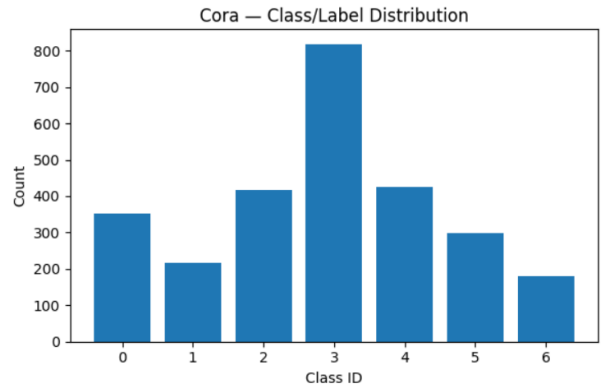
2.4 Exploratory plots

I include the following plots exported from my notebook:

- node degree distribution
- class distribution bar chart



(a) Node degree distribution.



(b) Class distribution.

Figure 1: Cora exploratory plots.

3 Experimental Setup

3.1 Fixed hyperparameters

I keep the following settings constant for all runs:

- hidden size: 64
- dropout: 0.5
- epochs: 200
- optimizer: Adam, learning rate 0.01, weight decay 5×10^{-4}
- loss: negative log likelihood on the train mask (model outputs log-softmax)

3.2 Input feature settings

I vary only the input feature dimensionality while keeping all other hyperparameters fixed:

- **Full features:** 1,433
- **AE features:** 64, 128, and 256

I train a small autoencoder once to obtain the compact features. I encode in **eval** mode with **no_grad** and reuse the saved embeddings for all models. The motivation is pre-hoc feature transformation to improve expressivity and to enable the LSTM aggregator without out-of-memory issues. On 1,433 dimensions I do not run the LSTM aggregator due to OOM.

3.3 Training and evaluation protocol

- depth is fixed at two layers for all models
- I treat the citation graph as undirected during message passing
- I select the epoch with the best validation accuracy and report test accuracy at that epoch (test@best-val)
- for each best configuration I also plot loss and accuracy curves, and report the confusion matrix and per-class precision, recall, and F1 on the test split

4 Architectures

4.1 GraphSAGE (two layers)

GraphSAGE forms node representations by aggregating neighbors and then applying a learnable transformation. For layer l and node v :

$$\mathbf{h}_v^{(l)} = \sigma\left(\mathbf{W}^{(l)} \text{CONCAT}\left(\mathbf{h}_v^{(l-1)}, \text{AGG}\{\mathbf{h}_u^{(l-1)} \mid u \in \mathcal{N}(v)\}\right)\right).$$

I evaluate three aggregators under identical settings:

- **mean:** elementwise average of neighbor features
- **max:** elementwise max-pool over neighbors
- **lstm:** an LSTM over a random permutation of neighbor features (used on AE features; not run on 1,433 due to OOM)

The hidden layer uses ReLU and dropout. The output layer applies log-softmax over 7 classes. The loss is computed only on train nodes.

4.2 Graph Attention Networks (two layers)

GAT introduces attention weights to score neighbors:

$$\mathbf{h}_v^{(l)} = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} \mathbf{W}^{(l)} \mathbf{h}_u^{(l-1)}\right),$$

where $\alpha_{vu}^{(l)}$ are learned, normalized attention coefficients. I vary the number of heads $H \in \{1, 2, 5\}$ while keeping all other settings fixed. The first layer uses H heads with concatenation before the nonlinearity (dimension $64 \times H$), and the final layer uses `concat=False` to produce class logits. The hidden layer uses ELU and dropout. The output applies log-softmax over 7 classes, and I train with NLL loss on the train mask.

5 Results

5.1 Results: Full Features (1433-D)

5.1.1 Summary across all configurations

Table 2: 1433-D features: all configurations with parameter counts and test accuracy (test@best-val).

Family	Variant	#Params	Test (%)
GraphSAGE	mean	184,391	80.8
GraphSAGE	max	184,391	78.5
GraphSAGE	lstm	— (OOM on 1433-D)	
GAT	1 head	92,373	81.4
GAT	2 heads	184,725	81.4
GAT	5 heads	461,781	80.3

5.1.2 Best GraphSAGE (mean): test metrics

Table 3: Per-class precision, recall, F1, and support on the test split.

Class ID	Precision	Recall	F1	Support
0	0.7417	0.6846	0.7120	130
1	0.7961	0.9011	0.8454	91
2	0.8973	0.9097	0.9034	144
3	0.8901	0.7618	0.8209	319
4	0.7853	0.8591	0.8205	149
5	0.7778	0.7476	0.7624	103
6	0.6042	0.9062	0.7250	64

Overall: accuracy = 0.8080, macro-avg $(P, R, F1) = (0.7846, 0.8243, 0.7985)$; test nodes = 1000.

Confusion matrix — GraphSAGE (mean)

Notes (GraphSAGE).

- Test accuracy: **80.8%**; parameters: **184,391**.
- Call out the *easiest* class (highest F1) and the *hardest* class (lowest F1) from the table above.

Table 4: Rows are true labels, columns are predicted.

	0	1	2	3	4	5	6
0	89	4	2	10	7	3	15
1	2	82	2	3	0	1	1
2	2	4	131	6	0	1	0
3	15	8	7	243	28	9	9
4	4	1	0	9	128	6	1
5	6	3	4	1	0	77	12
6	2	1	0	1	0	2	58

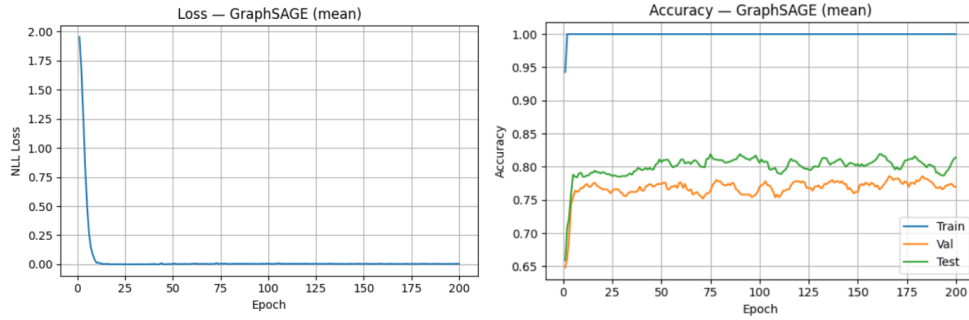


Figure 2: GraphSAGE (mean): loss and accuracy vs epochs (1433-D).

- Mention the dominant confusion pair(s) from the matrix (often $3 \leftrightarrow 4$ on Cora).

5.1.3 Best GAT (1 head): test metrics

Table 5: GAT (2 layers, 1 head): per-class precision, recall, F1, support on 1433-D.

Class ID	Precision	Recall	F1	Support
0	0.721	0.715	0.718	130
1	0.796	0.813	0.804	91
2	0.887	0.931	0.909	144
3	0.875	0.834	0.854	319
4	0.858	0.732	0.790	149
5	0.707	0.796	0.749	103
6	0.700	0.875	0.778	64

Overall: accuracy = 0.814, macro-avg $(P, R, F1) = (0.792, 0.814, 0.800)$; test nodes = 1000.

Table 6: GAT (1 head): confusion matrix on the test split (rows=true, cols=pred).

	0	1	2	3	4	5	6
0	93	5	3	9	3	8	9
1	4	74	5	7	0	1	0
2	1	1	134	6	1	1	0
3	11	7	5	266	14	11	5
4	10	3	0	14	109	11	2
5	5	3	4	1	0	82	8
6	5	0	0	1	0	2	56

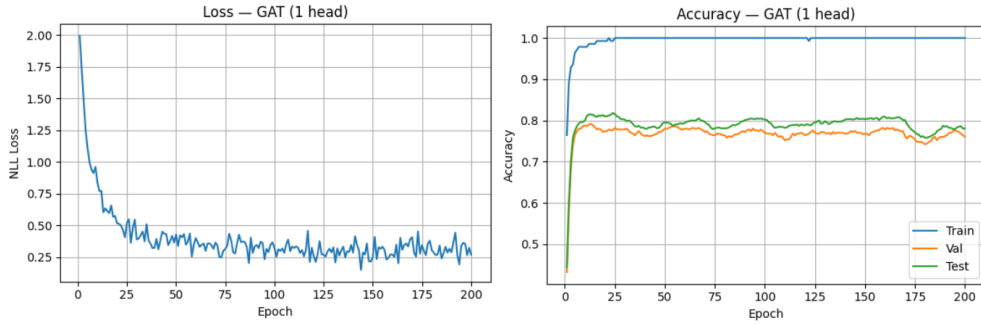


Figure 3: GAT (1 head): loss and accuracy vs epochs (1433-D).

Notes (GAT).

- Test accuracy: **81.4%**; parameters: **92,373**.
- Strongest class is **2** ($F1 \approx 0.91$). The main confusion is **3**↔**4**, which reduces class 4 recall (≈ 0.73).
- Compared to SAGE-mean, this model reaches similar accuracy with about *half* the parameters.

5.2 Results: AE Features (64-D)

5.2.1 Summary across all configurations

Table 7: 64-D features: all configurations with parameter counts and test accuracy (test@best-val).

Family	Variant	#Params	Test (%)
GraphSAGE	mean	9,159	75.9
GraphSAGE	max	9,159	73.1
GraphSAGE	lstm	75,719	58.4
GAT	1 head	4,757	75.3
GAT	2 heads	9,493	78.7
GAT	5 heads	23,701	77.0

5.2.2 Best GraphSAGE (mean): test metrics

Table 8: GraphSAGE (2 layers, mean): per-class precision, recall, F1, support on 64-D.

Class ID	Precision	Recall	F1	Support
0	0.5901	0.7308	0.6529	130
1	0.7400	0.8132	0.7749	91
2	0.8389	0.8681	0.8532	144
3	0.8759	0.7304	0.7966	319
4	0.7134	0.7517	0.7320	149
5	0.7812	0.7282	0.7538	103
6	0.6338	0.7031	0.6667	64

Overall: accuracy = 0.7590, macro-avg $(P, R, F1) = (0.7391, 0.7608, 0.7472)$; test nodes = 1000.

Table 9: Rows are true labels, columns are predicted.

	0	1	2	3	4	5	6
0	95	4	3	8	4	3	13
1	5	74	2	5	2	3	0
2	3	8	125	5	0	3	0
3	16	5	12	233	39	8	6
4	17	6	3	9	112	1	1
5	11	2	4	5	0	75	6
6	14	1	0	1	0	3	45

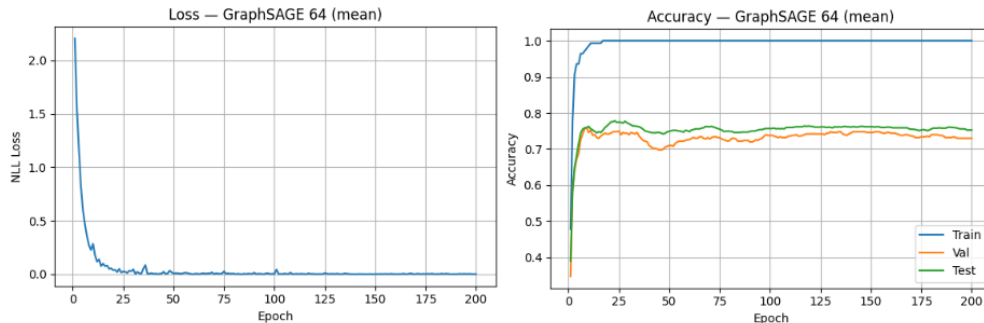


Figure 4: GraphSAGE (mean): loss and accuracy vs epochs (64-D).

Confusion matrix — GraphSAGE (mean)

Table 10: GAT (2 layers, 2 heads): per-class precision, recall, F1, support on 64-D.

Class ID	Precision	Recall	F1	Support
0	0.6562	0.8077	0.7241	130
1	0.7830	0.9121	0.8426	91
2	0.8442	0.9028	0.8725	144
3	0.9455	0.6520	0.7718	319
4	0.6961	0.8456	0.7636	149
5	0.8000	0.7767	0.7882	103
6	0.6835	0.8438	0.7552	64

Overall: accuracy = 0.7860, macro-avg $(P, R, F1) = (0.7727, 0.8201, 0.7883)$; test nodes = 1000.

Table 11: GAT (2 heads): confusion matrix on the test split (rows=true, cols=pred).

	0	1	2	3	4	5	6
0	105	4	4	4	6	0	7
1	4	83	1	3	0	0	0
2	2	4	130	3	1	3	1
3	18	10	14	208	48	13	8
4	14	3	1	2	126	2	1
5	10	1	4	0	0	80	8
6	7	1	0	0	0	2	54

5.2.3 Best GAT (2 heads): test metrics

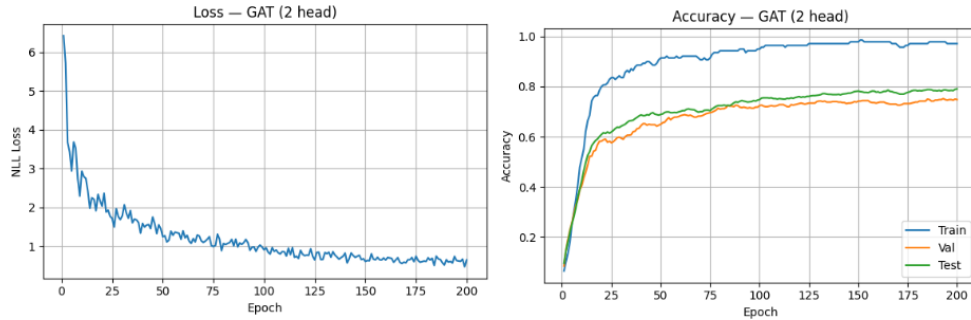


Figure 5: GAT (2 heads): loss and accuracy vs epochs (64-D).

5.3 Results: Autoencoder Features (128-D)

5.3.1 Summary across all configurations

5.3.2 Best GraphSAGE (mean): test metrics

Table 12: 128-D features: all configurations with parameter counts and test accuracy (test@best-val).

Family	Variant	#Params	Test (%)
GraphSAGE	mean	17,351	75.6
GraphSAGE	max	17,351	71.3
GraphSAGE	lstm	182,727	55.0
GAT	1 head	8,853	79.1
GAT	2 heads	17,685	79.4
GAT	5 heads	44,181	76.6

Table 13: Per-class precision, recall, F1, and support on the test split.

Class ID	Precision	Recall	F1	Support
0	0.6197	0.6769	0.6471	130
1	0.8734	0.7582	0.8118	91
2	0.8562	0.8681	0.8621	144
3	0.8382	0.7147	0.7716	319
4	0.7030	0.7785	0.7389	149
5	0.6864	0.7864	0.7330	103
6	0.6282	0.7656	0.6901	64

Overall: accuracy = 0.7560, macro-avg ($P, R, F1$) = (0.7436, 0.7641, 0.7506); test nodes = 1000.

Table 14: Rows are true labels, columns are predicted.

	0	1	2	3	4	5	6
0	88	4	4	15	6	6	7
1	9	69	4	4	3	1	1
2	3	1	125	10	0	4	1
3	17	4	8	228	38	14	10
4	9	1	2	10	116	8	3
5	8	0	3	3	1	81	7
6	8	0	0	2	1	4	49

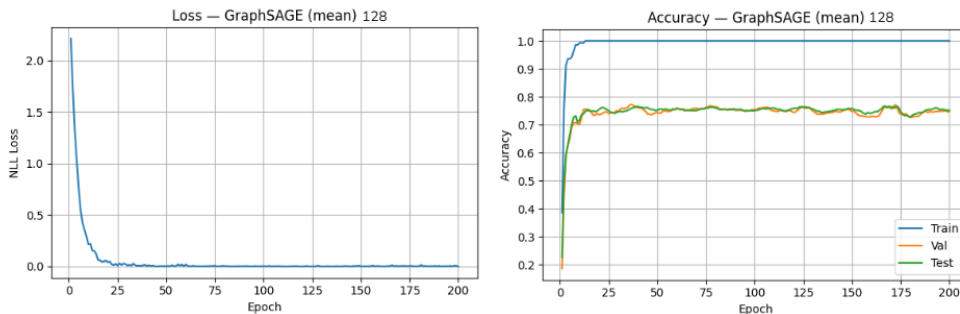


Figure 6: GraphSAGE (mean): loss and accuracy vs epochs (128-D).

Confusion matrix — GraphSAGE (mean)

Notes (GraphSAGE).

- Test accuracy: **75.6%**; parameters: **17,351**.

5.3.3 Best GAT (2 heads): test metrics

Table 15: GAT (2 heads): per-class precision, recall, F1, support on 128-D.

Class ID	Precision	Recall	F1	Support
0	0.7664	0.6308	0.6920	130
1	0.7921	0.8791	0.8333	91
2	0.8000	0.9444	0.8662	144
3	0.8700	0.7555	0.8087	319
4	0.7167	0.8658	0.7842	149
5	0.8020	0.7864	0.7941	103
6	0.7031	0.7031	0.7031	64

Overall: accuracy = 0.7940, macro-avg ($P, R, F1$) = (0.7786, 0.7950, 0.7831); test nodes = 1000.

Table 16: Rows are true labels, columns are predicted.

	0	1	2	3	4	5	6
0	82	7	7	13	9	7	5
1	0	80	5	5	1	0	0
2	1	3	136	2	2	0	0
3	3	6	18	241	36	7	8
4	3	2	0	12	129	3	0
5	8	2	4	1	1	81	6
6	10	1	0	3	2	3	45

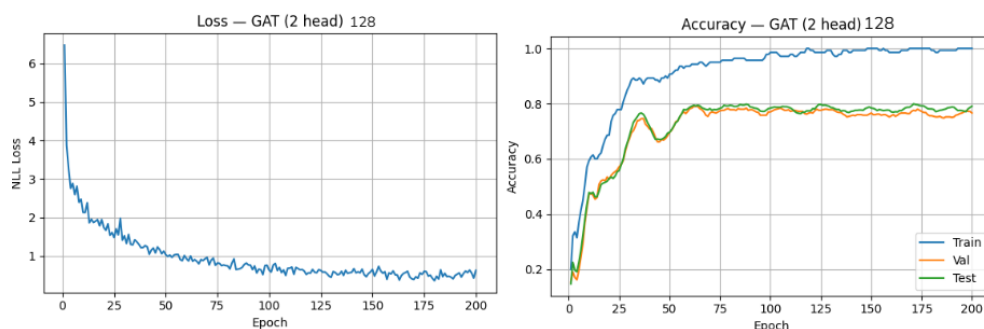


Figure 7: GAT (2 heads): loss and accuracy vs epochs (128-D).

Confusion matrix — GAT (2 heads)

Notes (GAT).

- Test accuracy: **79.4%**; parameters: **17,685**.

5.4 Results: Autoencoder Features (256-D)

5.4.1 Summary across all configurations

Table 17: 256-D features: all configurations with parameter counts and test accuracy (test@best-val).

Family	Variant	#Params	Test (%)
GraphSAGE	mean	33,735	74.8
GraphSAGE	max	33,735	74.4
GraphSAGE	lstm	593,351	57.1
GAT	1 head	17,045	79.3
GAT	2 heads	34,069	79.7
GAT	5 heads	85,141	79.3

5.4.2 Best GraphSAGE (mean): test metrics

Table 18: Per-class precision, recall, F1, and support on the test split (256-D).

Class ID	Precision	Recall	F1	Support
0	0.6612	0.6154	0.6375	130
1	0.8452	0.7802	0.8114	91
2	0.7399	0.8889	0.8076	144
3	0.8755	0.6614	0.7536	319
4	0.6597	0.8456	0.7412	149
5	0.7105	0.7864	0.7465	103
6	0.6711	0.7969	0.7286	64
<i>Overall:</i> accuracy = 0.7480, macro-avg ($P, R, F1$) = (0.7376, 0.7678, 0.7466); test nodes = 1000.				

Confusion matrix — GraphSAGE (mean)

Table 19: Rows are true labels, columns are predicted.

	0	1	2	3	4	5	6
0	80	3	6	11	14	8	8
1	6	71	6	4	4	0	0
2	4	4	128	6	2	0	0
3	10	4	28	211	40	18	8
4	11	1	2	4	126	4	1
5	5	1	2	5	1	81	8
6	5	0	1	0	4	3	51

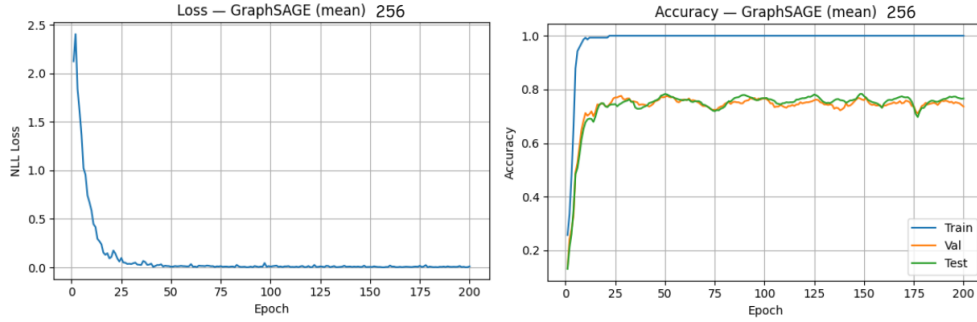


Figure 8: GraphSAGE (mean): loss and accuracy vs epochs (256-D).

Table 20: GAT (2 layers, 2 heads): per-class precision, recall, F1, support on 256-D.

Class ID	Precision	Recall	F1	Support
0	0.6618	0.6923	0.6767	130
1	0.7629	0.8132	0.7872	91
2	0.8313	0.9236	0.8750	144
3	0.8741	0.7837	0.8264	319
4	0.7834	0.8255	0.8039	149
5	0.8000	0.7767	0.7882	103
6	0.7500	0.7500	0.7500	64

Overall: accuracy = 0.7980, macro-avg ($P, R, F1$) = (0.7805, 0.7950, 0.7868); test nodes = 1000.

5.4.3 Best GAT (2 heads): test metrics

Table 21: Rows are true labels, columns are predicted.

	0	1	2	3	4	5	6
0	90	4	3	14	9	3	7
1	4	74	5	6	1	1	0
2	3	4	133	4	0	0	0
3	12	9	15	250	23	8	2
4	9	3	0	11	123	2	1
5	8	3	4	1	1	80	6
6	10	0	0	0	0	6	48

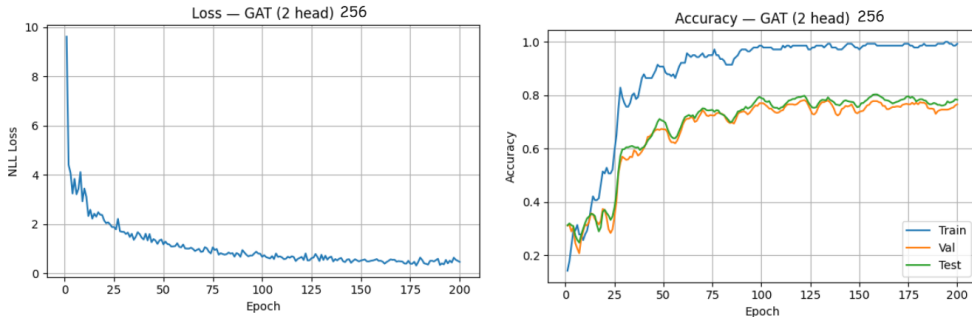


Figure 9: GAT (2 heads): loss and accuracy vs epochs (256-D).

Table 22: Combined test accuracy (%) across feature dimensions (test@best-val).

Family	Variant	1433-D	64-D	128-D	256-D
GraphSAGE	mean	80.8	75.9	75.6	74.8
	max	78.5	73.1	71.3	74.4
	lstm	— ¹	58.4	55.0	57.1
GAT	1 head	81.4	75.3	79.1	79.3
	2 heads	81.4	78.7	79.4	79.7
	5 heads	80.3	77.0	76.6	79.3

Confusion matrix — GAT (2 heads)

5.5 Analysis and Observations

Overall ranking (test@best-val)

- On full features (1433-D), GAT with 1–2 heads slightly edges GraphSAGE-mean (81.4% vs 80.8%).
- Under AE features (64/128/256-D), GAT with **2 heads** is consistently best (78.7/79.4/79.7%),

¹LSTM on 1433-D not run due to out-of-memory (OOM).

while GraphSAGE-mean trails (75.9/75.6/74.8%).

- LSTM aggregation underperforms even when enabled by AE (58.4/55.0/57.1% for 64/128/256-D) despite much higher parameter counts.

Parameter efficiency

- On 1433-D, **GAT-1h** reaches 81.4% with **92,373** params, about half of GraphSAGE-mean (80.8% at 184,391).
- With AE features, the best **GAT-2h** remains compact (e.g., 9,493 params at 78.7% on 64-D; 17,685 at 79.4% on 128-D; 34,069 at 79.7% on 256-D), offering strong accuracy per parameter.
- GraphSAGE-LSTM uses far more parameters (e.g., 75,719 at 64-D; 182,727 at 128-D; 593,351 at 256-D) but is not competitive.

Per-class behavior and confusion patterns (best models per family)

1433-D.

- **GraphSAGE-mean (80.8%)**: c2 is the easiest ($F1 \approx 0.90$). c6 shows very high recall (0.91) but low precision (0.60), i.e., more false positives into class 6. The largest confusion is **c3→c4 (28)**; c0→c6 (15) and c5→c6 (12) also appear.
- **GAT-1h (81.4%)**: c2 again strongest ($F1 \approx 0.91$). Class 4 is recall-limited (≈ 0.73) due to symmetric **c3↔c4 (14/14)** confusions. Class 6 keeps high recall (0.88) with moderate precision.

64-D.

- **SAGE-mean (75.9%)**: macro metrics drop vs 1433-D; c3 recall dips (more 3→4); c0 recall improves relative to precision (more conservative predictions for some classes).
- **GAT-2h (78.7%)**: better macro recall; c1/c2/c4/c6 are strong; c3 remains the main source of errors (3→4 and some spread), but overall balance is improved versus SAGE.

128-D.

- **SAGE-mean (75.6%)**: c1 has high precision with lower recall (more conservative on class 1). c3 recall remains lower than precision.
- **GAT-2h (79.4%)**: c2 recall is very high (≈ 0.94); precision/recall are well balanced across most classes, yielding a stable macro-F1.

256-D.

- **SAGE-mean (74.8%)**: accuracy decreases slightly vs 128-D; c3 recall (≈ 0.66) and persistent 3→4 confusions dominate the errors.
- **GAT-2h (79.7%)**: strong on c2/c4 and solid on c3; class-wise metrics are balanced, keeping performance near 80%.

Feature dimension effects

- Compressing to 64/128/256-D hurts GraphSAGE more than GAT: SAGE-mean drops from 80.8% (1433-D) to 75–76% on AE variants, whereas GAT with 2 heads stays near 79–80%.
- The AE enables LSTM but does not improve it; order-sensitive aggregation with limited labels and stochastic neighborhoods likely contributes to weaker generalization.

Training behavior (curves)

- Train accuracy quickly saturates near 1.0 across settings (small train split), so validation curves are the reliable signal. Selecting test@best-val avoids last-epoch drift.
- Validation curves for the best GAT models are stable; SAGE shows mild variance on AE features but remains monotonic enough for clear model selection.

Practical takeaways

- For a fixed 2-layer design on Cora, **GAT with 1–2 heads** is a strong, parameter-efficient default on both full and AE features.
- **GraphSAGE-mean** remains a solid baseline on full features but loses more under AE compression.
- The dominant error mode across settings is the **c3 vs c4** boundary; improving features for these topics or adding regularization/priors that separate them would likely yield the biggest gains.

6 t-SNE Analysis: Expressiveness and Oversmoothing

6.1 GAT (2 layers, 1 head, full 1433-D)

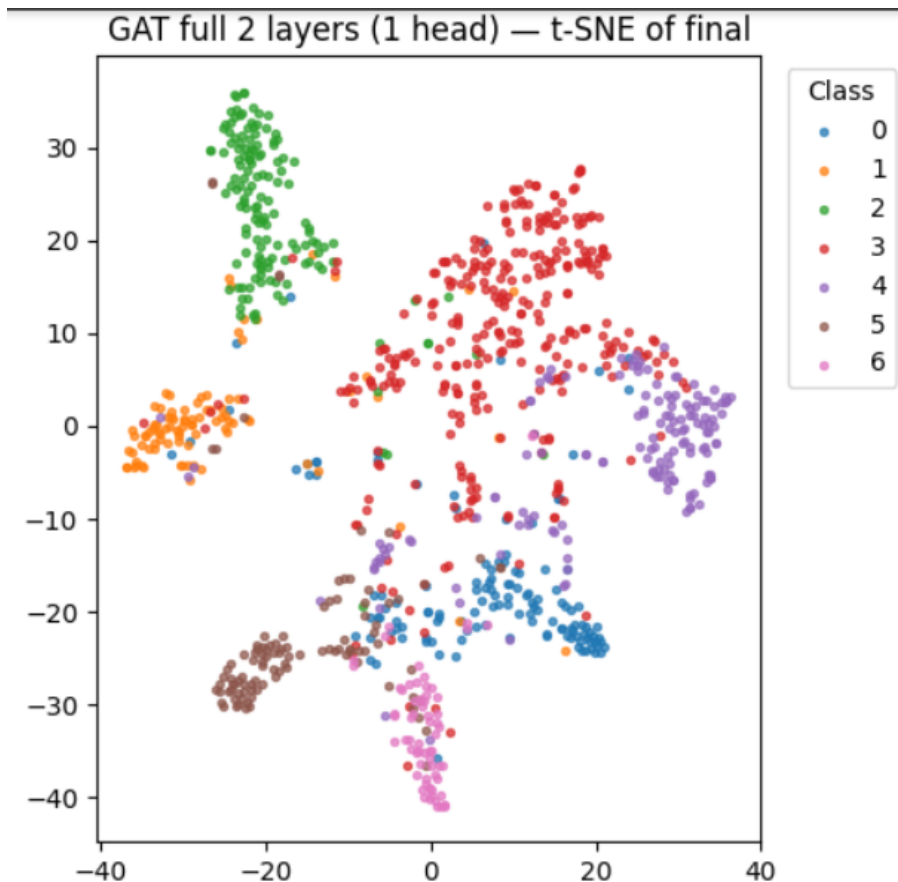


Figure 10: GAT (2 layers, 1 head): t-SNE of final-layer embeddings (test nodes).

Observations

- Clear, compact clusters for several classes (e.g., class 2), consistent with its high F1 in the results.
- The class-3 vs class-4 boundary shows visible overlap, matching the $3 \leftrightarrow 4$ confusions and lower recall for class 4.
- Class 6 forms a cluster but sits near tails of other classes, aligning with high recall but moderate precision.

6.2 GAT (deep, ~ 19 layers, 1 head, full 1433-D)

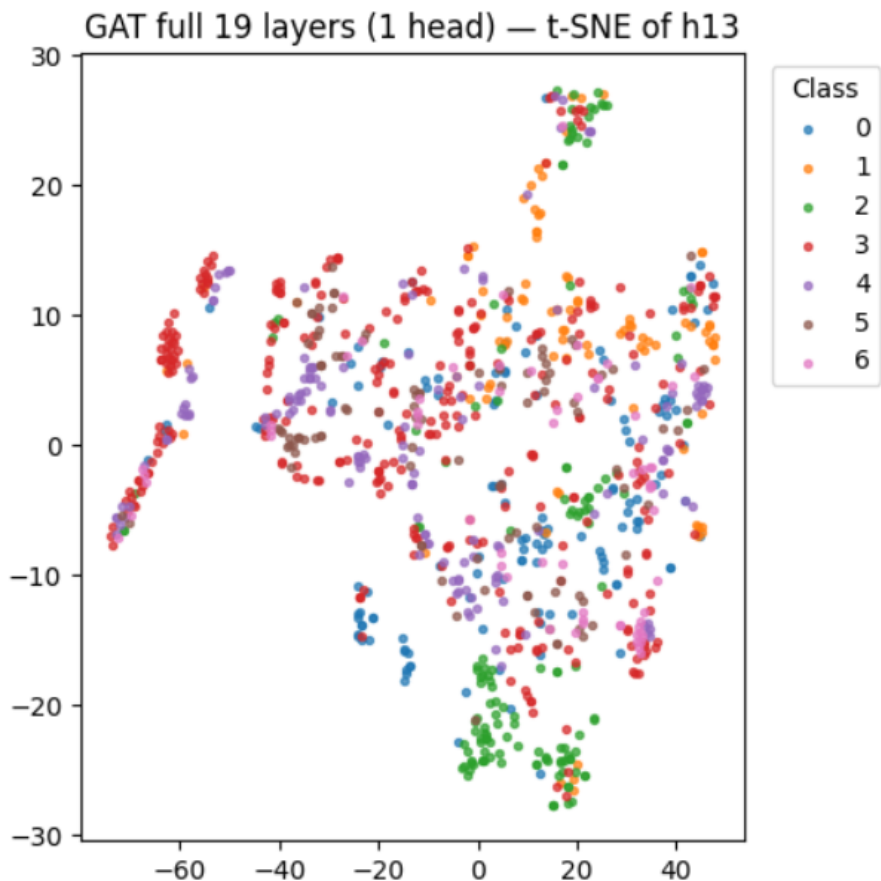


Figure 11: GAT (deep, ~ 19 layers, 1 head): t-SNE of a late hidden layer (test nodes).

Observations

- Widespread color mixing and loss of compact class structure indicate *oversmoothing*.
- After many hops, embeddings become similar across nodes, blurring decision boundaries; this aligns with weaker separability than the 2-layer model.
- Attention with a single head does not prevent smoothing at this depth; shallow GAT retains better class contrast on Cora.